Language Identification From Audio

Rachanaa Raghuthama Malamandi

Abstract

This document describes our solution for the project including the used methods, experiments and their evaluation.

1 Task 1: Reproduce and Improve the Baseline

To reproduce and improve the baseline we conducted some tests on different models, with tuning the following hyperparameters:

1. training epochs
2. learning rate
3. learning rate schedule
4. warmup ratio
5. weight decay

Our received results with regards to the evaluation set accuracy is displayed in Figure 1. The models we compared against each other are thereby facebook/mms-300m (Pratap et al., 2023), utter-project/mHuBERT-147 (Boito et al., 2024) and facebook/wav2vec2-xls-r-300m (Babu et al., 2021). We thereby choose the cosine learning rate decay, as it gives us a boost in learning the first few epochs before the learning rate declines more sharply. This allows us to adapt the models fast on the new task, before we reduce the learning rate to converge.

2 Task 2: Address Dataset Bias

The dataset used in the project consisted of only 5 speakers per language. Since it is highly likely, that the model would implicitly learn to associate the language with the speakers, which provided the audio samples of that language, we need to find a way to reduce this speaker-bias. To counteract this, we used a *Domain-Adversarial Neural Network* (DANN) (Ganin et al., 2016). DANNs consists of two heads. One for the results of the classification task itself and another one to classify a domain label. In the backpropagation, the loss of our classification task is normally backpropagated. But we use a special layer called *gradient reversal layer* for the domain-specific classification loss, which multiplies this loss with a negative constant. By doing that, we disincentivize the neural network to find the correct domain label and thus obscure the domain-specific knowledge. A graphical representation of this process can be seen in Figure 2.

In our concrete case, we want to classify the spoken language with the domain of different speakers. This means, that we try to obscure the speaker identity as much as possible, while still wanting to predict the correct language, when we use the DANN.

To control the strength of the gradient reversal layer of the DANN, we use a technique called *dynamic alpha*. We thereby use the parameter α to dynamically decide the value of the negative constant of the gradient reversal layer. We start thereby with a value of zero to increase the learning for the overall classification problem. We then increase α exponentially with regards to the training step, causing our model to focus more and more on maximizing the domain-specific loss, which in turn minimizes the model loss more and more.

Additionally to using a DANN model, we use *random cropping* (Boigne, 2021) as a data augmentation technique. Random cropping selects random snippets of fixed size from audio samples. This is done to prevent the model from overfitting to specific samples or audio sample starts.

3 Task 3: Analyze the Model

In this section, we discuss different problems mentioned in the project description and discuss the results of our model compared to the baseline. We start with discussing the bias existent in the dataset. This bias mainly comes from having multiple au-

	Test 1	Test 2	Test 3
Model	facebook/mms-300m	utter-project/mHuBERT-147	Facebook/wav2vec2-xls-r-300m
Dropout	0.1	0.1	0.1
Epochs	10	15	15
Learning rate	3E-05	1E-04	1E-04
lr_sceduler_type	Cosine	Cosine	Cosine
warmup_ratio	0.15	0.15	0.1
weight_decay	0.1	0.15	0.1
Accuracy	27%	53%	22%

Figure 1: Test results for different models and hyperparameters

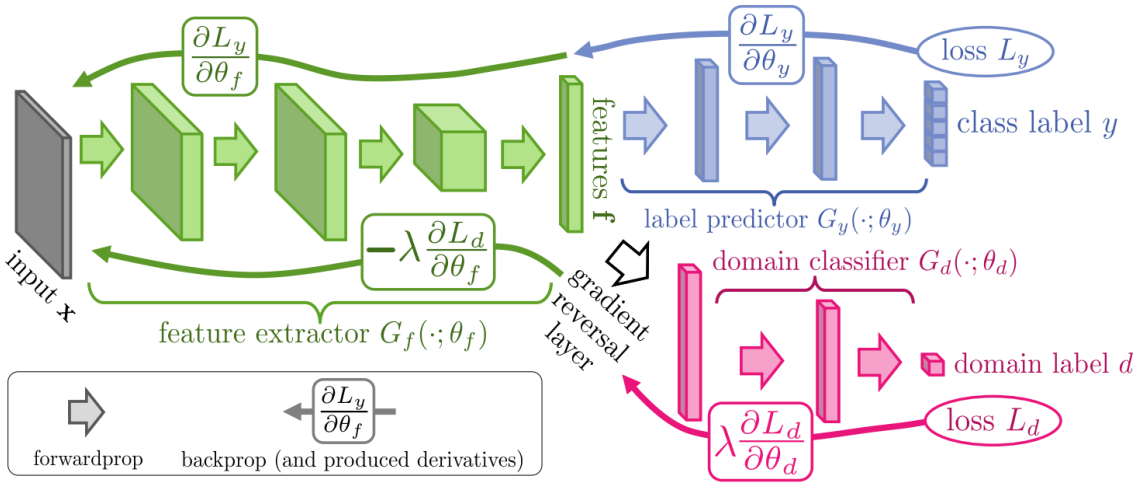

Figure 2: Inner workings of a DANN model (Ganin et al., 2016)

dio recordings from the exact same people in the dataset. A person's unique voice is easier for a computer to recognize than complex language rules. Because of this, the neural network takes the lazy route. It simply memorizes who is speaking instead of learning what language they are speaking.

When the model gets an audio from a completely new speaker, it completely fails because it only learned the speaker and not the language. To fix the bias issue, a Domain-Adversarial Neural Network (DANN) was used. This includes a second head which predicts based only on the speaker. However, the trick is in the Gradient Reversal Layer. This layer inverts the gradients forces the main feature extractor to actively forget the speaker, which is exactly what we want to achieve. The solution was moderately successful. The model reached $\tilde{6}9\%$ accuracy and the t-SNE plots speaks for it. However, looking at the Bias Summary displayed in Figure 3, the difference between the max_accuracy and min_accuracy is large for some languages, showing that some residual speaker bias still remains.

The model heavily confuses languages that share linguistic features. Major confusion is with Hindi and Urdu as they are virtually the same conversational language: True Hindi was misclassified as Urdu 54 times. True Urdu was misclassified as Hindi 11 times. Similarly, Marathi and Konkani: True Marathi was misclassified as Konkani 67 times. True Konkani was misclassified as Marathi

--- Bias summary ---

Language	Min_Accuracy	Max_Accuracy
assamese	40.0%	100.0%
bengali	0.0%	100.0%
bodo	20.0%	100.0%
dogri	0.0%	100.0%
gujarati	20.0%	100.0%
hindi	0.0%	100.0%
kannada	33.33%	100.0%
kashmiri	60.0%	100.0%
konkani	0.0%	100.0%
maithili	0.0%	100.0%
malayalam	0.0%	100.0%
manipuri	60.0%	100.0%
marathi	0.0%	100.0%
nepali	0.0%	100.0%
odia	0.0%	100.0%
punjabi	0.0%	100.0%
sanskrit	40.0%	100.0%
santali	20.0%	100.0%
sindhi	20.0%	100.0%
tamil	40.0%	100.0%
telugu	33.33%	100.0%
urdu	20.0%	100.0%

Figure 3: Bias Summary of the DANN model of evaluation set

43 times. Another such case is Punjabi and Dogri: True Punjabi was misclassified as Dogri 42 times. True Dogri was misclassified as Punjabi 36 times as they sound similar and contain similar vocabularies.

The comparison of the classification results for the evaluation set of both models is shown in Figure 4. We can see there, that there seems to be a significant improvement in the classification of each language in the DANN model compared to the base model. This improvement can also be seen in the t-SNE plots of the evaluation set of both models displayed in Figure 5. The t-SNE visualizations of the last-layer are colored by speaker identity, allowing us to assess whether the model's embeddings are organized around speaker characteristics.

In the baseline model, the plot is diffused and unstructured, with no clear compact clusters indicating the model has not strongly encoded speaker identity, but also has not learned the language. In the DANN model, the plot forms approximately 20 tight and compact clusters and within each cluster, points of many different speaker colors are thoroughly mixed together. This color mixing within clusters indicates that the DANN representations are not organized by speaker identity and the clusters probably correspond to the different language groups. The Gradient Reversal Layer thus seems to have successfully penalized the encoder for encoding speaker features, resulting in representations that assumingly group utterances by language regardless of who the speaker is. This confirms that the DANN model most likely reduces speaker identity encoding and learns a more language-centric representation compared to the baseline.

References

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Per Language Accuracy
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	precision	recall	f1-score	support
konkani	0.1204	0.3067	0.1729	150
hindi	0.0958	0.1067	0.1009	150
sanskrit	0.5469	0.4667	0.5036	150
santali	0.3160	0.6067	0.4155	150
marathi	0.2432	0.1200	0.1607	150
sindhi	0.1818	0.1733	0.1775	150
kashmiri	0.4051	0.7400	0.5236	150
bodo	0.4019	0.2867	0.3346	150
tamil	0.6571	0.1533	0.2486	150
maithili	0.3559	0.1400	0.2010	150
kannada	0.3286	0.1533	0.2091	150
telugu	0.2547	0.3600	0.2983	150
punjabi	0.1525	0.0600	0.0861	150
malayalam	0.4545	0.2000	0.2778	150
nepali	0.2797	0.5800	0.3774	150
dogri	0.2205	0.3867	0.2809	150
manipuri	0.3836	0.5933	0.4660	150
assamese	0.5000	0.3000	0.3750	150
urdu	0.3525	0.2867	0.3162	150
gujarati	0.2481	0.2200	0.2332	150
odia	0.3448	0.0667	0.1117	150
bengali	0.1429	0.0533	0.0777	150
accuracy			0.2891	3300
macro avg	0.3176	0.2891	0.2704	3300
weighted avg	0.3176	0.2891	0.2704	3300

(a). Classification Report of the base model

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--- Classification report ---
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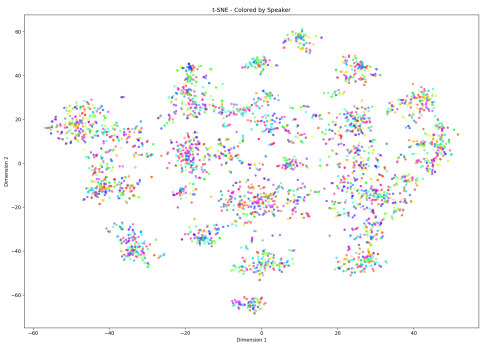
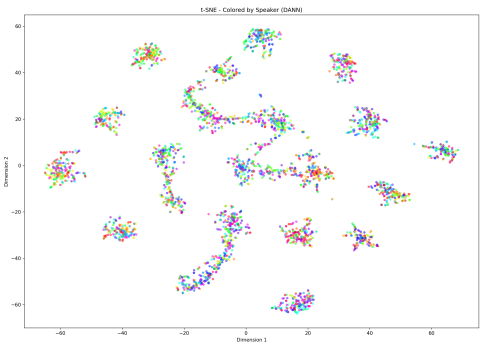
	precision	recall	f1-score	support
konkani	0.35	0.44	0.39	150
hindi	0.63	0.15	0.24	150
sanskrit	0.90	0.79	0.84	150
santali	0.80	0.68	0.73	150
marathi	0.43	0.54	0.48	150
sindhi	0.52	0.75	0.61	150
kashmiri	0.87	0.88	0.87	150
bodo	0.85	0.74	0.79	150
tamil	0.86	0.84	0.85	150
maithili	0.60	0.43	0.50	150
kannada	0.71	0.85	0.78	150
telugu	0.85	0.81	0.83	150
punjabi	0.57	0.57	0.57	150
malayalam	0.88	0.89	0.89	150
nepali	0.86	0.75	0.80	150
dogri	0.57	0.51	0.54	150
manipuri	0.67	0.96	0.79	150
assamese	0.87	0.91	0.89	150
urdu	0.59	0.69	0.63	150
gujarati	0.64	0.76	0.70	150
odia	0.75	0.57	0.65	150
bengali	0.72	0.68	0.70	150
accuracy			0.69	3300
macro avg	0.70	0.69	0.69	3300
weighted avg	0.70	0.69	0.69	3300

(b). Classification report of the DANN model

Figure 4: Comparison of the Classification reports between the base model and the DANN model



(a). t-SNE plot for the baseline model



(b). t-SNE plot for the DANN model

Figure 5: Comparison of t-SNE graphs between the base model and the DANN model